Continuous Development and Maintenance of an Operational Safety Critical System

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Abstract

The standard development lifecycle for railway projects typically assumes a linear progression from concept through definition, implementation, deployment, and maintenance, often spanning multiple years, with established standards and processes reflecting this approach. However, modern railway systems operate in an environment where functionality must evolve continuously while maintaining safe, uninterrupted operations.

This paper examines a case study of a mission-critical, 24/7 automated freight system, which requires not only ongoing maintenance but also the frequent introduction of new and modified functionalities. The continuous evolution of such systems challenges traditional lifecycle assumptions and necessitates the adaptation of established safety and engineering processes.

Key challenges are identified, including maintaining safety assurance under compressed deployment timelines, managing frequent system updates, and ensuring robust configuration and integration practices within an operational environment. The paper presents practical insights and lessons learned from implementing multiple safe deployments per year, demonstrating how tailored processes and strong stakeholder collaboration can enable effective, efficient, and safe delivery of system updates in a continuous development context, based on tailoring the standard processes, using key concepts from other methodologies, such as agile, lean and evolutionary prototyping.

Keywords: *Safety Systems, Maintenance, Railway, Operational Safety, Agile, Lean, evolutionary prototyping EN50126.*

1 Introduction

The standard development lifecycle for a railway project assumes a project starts from concept, through to definition, implementation, rollout and maintenance and usually runs across multiple years.

But what happens after you have rolled out your new system?

What if you are continually refining the system and introducing new safety features?

What if you are doing this in short timeframes?

What if the system is mission critical and cannot afford downtime?

How can you use the normal standards and processes to support this?

In this paper, we examine a case study facing exactly these questions.

The key benefits of this paper are:

1. Advancement of knowledge of applying multiple methodologies within the same development lifecycle for a railway project.
2. Improved understanding of how to effectively and efficiently tailor the standard development lifecycle for a railway project in practice.
3. Sharing experience from use of such a tailored development lifecycle in an industrial case study.

2 Acronyms, Abbreviations, and Definitions

2.1 Acronyms and Abbreviations

Acronym	Description
Comms.	Communications System
ISA	Independent Safety Assessor
RAMS	Reliability Availability Maintainability & Safety
SFAIRP	So Far As Is Reasonably Practicable
SIL	Safety Integrity Level

Table 1:Acronym Table

2.2 Definitions

Following are the definitions useful for this paper;

2.2.1 Hazard

Condition that could lead to an accident.

2.2.2 Agile

A development approach that delivers products which changes in small, iterative increments, with continuous stakeholder feedback, frequent verification, and rapid adaptation to changing requirements while maintaining safety assurance.

2.2.3 Lean

A management philosophy focused on maximizing customer value by eliminating waste, reducing rework, improving workflow efficiency, and continuously improving engineering and assurance processes throughout the product lifecycle.

2.2.4 Evolutionary Prototyping

A software and systems engineering approach where an initial prototype is developed, evaluated by users, and then continuously refined and expanded until it becomes the final operational system.

2.2.5 System

Set of interrelated elements considered in a defined context as a whole and separated from their Environment.

3 Literature Survey

Historically, railway signalling and control systems have been developed using structured lifecycle models that provide a systematic framework for requirements development, system design, implementation, verification, validation and safety assurance. Standards such as EN50126, EN50128 and EN50129 were developed to support the safe delivery and operation of safety-critical railway systems. These standards generally assume a lifecycle that progresses from concept through implementation and commissioning before entering an operational maintenance phase. However, modern railway systems increasingly require functional enhancements (due to flexibility allowed by modern railway technology), corrective changes and operational improvements long after commissioning, creating a need for approaches that can support controlled change within operational systems while maintaining safety assurance.

The challenge of managing continuous change has been extensively studied within the software and systems engineering literature. Agile Software Development emerged as a response to the limitations of traditional plan-driven approaches such as the Waterfall and V-Models, particularly in environments where requirements are uncertain or evolve over time. The Agile Manifesto defines Agile through the values of customer collaboration, responsiveness to change, continuous delivery of value and the importance of people and interactions (Beck et al., 2001; PMI, 2022). Agile is therefore widely recognised as a philosophy based on values and principles rather than a single development methodology.

The literature consistently identifies the ten Agile principles as the foundation of Agile practice. These principles seek to improve responsiveness, customer satisfaction and product quality through iterative development, early feedback and continuous learning (Beck et al., 2001). Research further reports benefits including increased flexibility, improved stakeholder collaboration, enhanced transparency, earlier defect identification, reduced rework and improved responsiveness to changing customer needs (Dybå & Dingsøyr, 2008; Temole & Atanasova, 2025; Anifa et al., 2024). The literature also supports the claim that Agile approaches require careful tailoring in highly regulated and safety-critical environments where documentation, traceability, governance and formal verification remain necessary (Hoy & Zu, 2023).

For the continuous development of operational railway systems, the Agile literature supports the following principles as being particularly relevant to Continuous Safety-Critical Development:

1. Customer collaboration and stakeholder engagement (Beck et al., 2001; PMI, 2022).
2. Responsiveness to changing requirements and operational needs (Beck et al., 2001; PMI, 2022).
3. Early and continuous delivery of valuable functionality (Beck et al., 2001).
4. Frequent delivery cycles and incremental enhancement (Beck et al., 2001).
5. Early feedback and continuous learning (Beck et al., 2001).
6. Close collaboration between business stakeholders and development teams (Beck et al., 2001).
7. Self-organising and collaborative teams (Beck et al., 2001).
8. Technical excellence and engineering discipline (Beck et al., 2001).
9. Continuous testing and early defect identification (Temole & Atanasova, 2025).
10. Continuous improvement through regular reflection and adaptation (Beck et al., 2001).

These Agile principles are particularly relevant to operational railway systems where modifications and enhancements must be introduced frequently without interrupting safe operation. The literature suggests that successful implementation requires balancing agility and responsiveness with the governance, traceability and assurance obligations associated with safety-critical systems (Hoy & Zu, 2023; Temole & Atanasova, 2025).

Lean Management provides a complementary perspective for supporting continuous development and operational improvement. Lean originated from the Toyota Production System and was developed in an environment characterised by scarce resources and the need to maximise effectiveness (Ohno, 1988). Over time, Lean evolved into a broader management philosophy based on customer value, waste reduction and continuous improvement (Womack et al., 1990). The literature emphasises that Lean should not be viewed simply as a collection of tools, but as a holistic approach influencing organisational culture, decision-making and process effectiveness (Womack & Jones, 1996; Seddon, 2005).

A fundamental concept within Lean literature is the distinction between process effectiveness and resource effectiveness. Rather than maximising the utilisation of individual resources, Lean focuses on improving the flow of work and the efficient delivery of value to the customer (Womack & Jones, 1996). Activities that generate delays, waiting time, excessive handovers, unnecessary variation or rework are considered waste because they reduce process performance. Consequently, Lean seeks to minimise cycle times, improve workflow consistency, reduce bottlenecks and enhance organisational effectiveness (Ohno, 1988; Seddon, 2005).

The literature also highlights the importance of customer focus, early communication and collaboration. Studies demonstrate that inadequately defined requirements, poor stakeholder engagement and weak communication interfaces are major causes of inefficiency, rework and project delays (Womack & Jones, 1996; Seddon, 2005). In engineering and safety-critical environments, effective collaboration and communication are therefore regarded as essential mechanisms for reducing waste and improving outcomes. Lean literature further emphasises the importance of continuous improvement and cross-functional cooperation in managing complex project environments (Womack et al., 1990; Seddon, 2005).

For the continuous development and maintenance of operational railway systems, the Lean literature supports the following principles:

1. Maximising customer value (Womack & Jones, 1996).
2. Eliminating waste (Ohno, 1988; Womack & Jones, 1996).
3. Minimising rework (Seddon, 2005; Womack & Jones, 1996).
4. Reducing bottlenecks (Ohno, 1988; Seddon, 2005).
5. Reducing unnecessary variation (Ohno, 1988; Seddon, 2005).
6. Eliminating non-value-adding activities (Womack et al., 1990; Womack & Jones, 1996).
7. Optimising process flow and process effectiveness (Womack & Jones, 1996; Seddon, 2005).
8. Preventing defects and misunderstandings through early communication (Womack & Jones, 1996; Seddon, 2005).
9. Continuous improvement (Ohno, 1988; Womack et al., 1990).
10. Cross-functional collaboration and interface management (Womack et al., 1990; Seddon, 2005).

The literature describes Evolutionary Prototyping as a way of developing software where an early version, or prototype, of the product is built quickly, then continuously refined and expanded based on user feedback until it evolves into the final system.

Evolutionary prototyping focuses on gathering a correct and consistent set of requirements. The process lends strength to building quality software by means of the ongoing clarification of existing requirements and the discovery of previously missing or unknown requirements (Ryan et al., 2001).

(Morgan, 2022) defines the four key steps involved with the Evolutionary Prototyping or Rapid Prototyping model are:

1. Initial Concept: Gather initial requirements
2. Create Prototype: Refine requirements
3. Refine Prototyping: Describe all requirements in full detail while creating each milestone
4. Release final product: Develop a final solution based on prototype development.

(S. Hekmatpour et al, 1987) argue that evolutionary prototyping can be regarded as a software system that is continually evolving entity rather than as the product of a series of developmental phases followed by a maintenance phase.

Evolutionary Prototyping literature emphasises the continuous user involvement in evolutionary prototyping. (Sommerville, 2007) notes that a key principle of evolutionary prototyping is continuous user involvement, where users regularly evaluate successive prototype versions and provide feedback that drives system refinement and requirements evolution.

(SEBoK Editorial Board, 2025) states that incremental development delivers a system in manageable increments and provides stakeholders with early, partial versions of the system for verification, validation, and feedback before the final system is complete.

For the continuous development and maintenance of operational railway systems, the Evolutionary Prototyping literature supports the following principles:

1. Continuous Evolution (Nothing is Thrown Away): The initial working model forms the core foundation of the final system, meaning code and design elements are steadily built upon rather than discarded (S. Hekmatpour et al, 1987).
2. Iterative Development in repetitive cycles of building, testing, evaluating, and adjusting features based on real-world usage (SEBoK Editorial Board, 2025).
3. Active User Involvement where the stakeholders and end-users test functional increments early and often, directly guiding the direction of the next development phase (Sommerville, 2007).
4. Adaptive Requirements that are not assumed to be fully known at the start, but instead, emerge and mature as stakeholders interact with the prototype (Morgan, 2022; Ryan et al., 2001).
5. Incremental Delivery: Functionality is added in gradual steps, delivering usable increments of the system to production or evaluation environments incrementally (SEBoK Editorial Board, 2025).

(Ryan et al, 2001) notes that the strengths of the evolutionary prototyping are also potentially areas of special concern, such as management of evolutionary prototyping is a major contributor to the building of 'good' software but requires strong control, a major benefit of evolutionary prototyping for a process model is that it leads to systems that meet known user requirements and yet can change to meet the user requirements that are discovered later, but scope creep needs to be carefully managed.

(Sommerville, 2007) contrasts evolutionary prototyping where the objective of the process is to work with the customer to explore their requirements and deliver a final system. The development starts with the parts of the system that are understood. The system evolves by adding new features proposed by the customer, with throwaway prototyping, where the objective of the evolutionary development process is to understand the customer's requirements and hence develop a better requirements definition for the system. The throwaway prototype concentrates on experimenting with the customer requirements that are poorly understood.

In summary, the literature notes the key aspects of evolutionary prototyping that emphasize building initial software to develop and refine customer requirements, with the final software solution being incrementally built upon the enhanced prototype.

Both Agile and Lean literature emphasise the importance of stakeholder engagement, collaboration, early feedback and continuous improvement. Agile focuses primarily on managing change and delivering value through iterative development and frequent feedback cycles, while Lean focuses on improving process effectiveness through waste reduction, improved communication and optimisation of workflow. Evolutionary Prototyping complements these approaches

by supporting the progressive refinement of system functionality through iterative development, continuous stakeholder involvement and ongoing evaluation of evolving system increments. Through repeated cycles of development, testing, user assessment and feedback, requirements are clarified, functionality is refined and confidence in the evolving solution is progressively increased. In railway applications, these evaluation activities may include simulation-based testing, integration laboratories, operational trials, staged commissioning and user-focused validation exercises that provide feedback before wider operational deployment. Together, these approaches provide complementary mechanisms for supporting the continuous evolution of operational systems while maintaining quality, efficiency, stakeholder confidence and alignment with operational needs (Beck et al., 2001; Womack & Jones, 1996; Seddon, 2005; Sommerville, 2007; SEBoK Editorial Board, 2025).

The reviewed literature therefore suggests that the successful continuous development and maintenance of operational safety-critical railway systems depend upon balancing the discipline of traditional assurance processes with the flexibility and process focus provided by Agile and Lean principles, together with the iterative refinement, active stakeholder involvement and continuous evaluation inherent in Evolutionary Prototyping. In particular, the Agile principles of continuous stakeholder engagement, frequent, iterative delivery, continuous testing, technical excellence and continuous improvement, together with the Lean principles of customer value, waste reduction, process optimisation, incremental development, ongoing evaluation, optimisation and cross-functional collaboration, and the Evolutionary Prototyping principles of continuous evolution, active user involvement, incremental delivery and continuous evaluation, provide the theoretical basis for the deployment practices discussed in Section 5, including early engagement with the Operator, strong configuration management, rigorous verification activities, integrated safety analysis, laboratory simulation, extensive validation, controlled site testing, open engagement with the Independent Safety Assessor, collaborative teamwork and proactive workload management across multiple annual deployments. The recurring evaluation of functionality throughout development and deployment helps ensure that system enhancements remain aligned with operational requirements while maintaining safety and stakeholder confidence.

Collectively, these principles support the controlled introduction of change into operational safety-critical railway systems through continuous feedback, structured validation, collaborative decision-making and progressive enhancement while maintaining the assurance, traceability and safety obligations required by railway standards.

4 Case Study

The case study involves a mission critical, 24 x 7, automated, live operation freight operation where Hitachi rail is the integrator. There is a reliance on SIL rated automated safety systems underpinning not just maintenance updates but the continuous introduction of novel and changed functions, across multiple (up to 8 drops) planned deployments a year. A holistic approach is used to ensure that the system remains safe and operational throughout the changes, and to minimise down time. Whilst there is a safety-first approach, it is important to note that attention is also given to operational impacts across the lifecycle, as availability and reliability are equally important aspects of the project.

The case study shows a strong alignment with the Agile, Lean and Evolutionary Prototyping concepts summarised in the Literature Survey. Rather than adopting Agile practices mechanically, the case study shows that the project has tailored traditional railway assurance processes to achieve the underlying objectives of Agile and Lean while maintaining compliance with safety-critical railway standards.

The case study also demonstrates a practical implementation of the principal ideas of Evolutionary Prototyping within a safety-critical railway environment, that instead of treating the system as "complete" after commissioning, it is managed as a continuously evolving operational system. A key difference is that traditional evolutionary prototyping literature is often focused on managing uncertain requirements during software development, whereas this case study applies the same principles in a highly regulated SIL-rated operational railway system. Consequently, evolutionary prototyping is combined with formal safety assurance, hazard management, configuration management, and ISA engagement to maintain safety while continuously evolving the system.

4.1 Early and Continuous Engagement with Operator

Planning of the project relies on early engagement with the Operator to determine and maintain the overall roadmap of enhancements each year. There are multiple deployments planned per year (up to 8 drops), and each has agreed list of changes that are mutually agreed in advance (up to a year ahead). For each of the proposed changes there is a combined (Hitachi and Operator) detailed review, including identifying any requirements impact, architecture impact, safety impact, and testing impact, etc. There is a strong focus on human factors and impact assessment from the Operator and as the changes are progressed, there is early exposure for the Operator, to updates in the testing labs and for controlled site testing. This strong collaborative approach ensures that the requirements associated with a change are established early on in the process, then continuously evolved and adapted, at each stage, such that the delivered requirements are correctly aligned with the implementation of the change and well understood by the Operator.

4.2 Strong Configuration Management

Strong configuration management is essential. Detailed management of agreed updates for each deployment is maintained, including sign off (from all applicable stakeholders) for each version. There is strong control of supported hardware and software baselines, supported by built in engineering checks of approved baselines and the deployment and commissioning processes mandate rigorous cross-checking and signoff of installed updates.

4.3 Verification, Verification, Verification

Verification checks are built in at each step of the lifecycle. This is critical, as if the requirements are ambiguous or unclear at the start, there is no time to correct later in the lifecycle.

Processes build in the verification, from the concept (identification of a proposed change) to allocation to a planned deployment (analysis of implementation, testing and safety impacts), to implementation, safety analysis, testing and deployment.

This level of verification requires strong, early and detailed communication between subsystems and system level teams to ensure that designs are clear and will work correctly when put together.

4.4 Safety Analysis

The traditional analysis (for example starting with a Preliminary Hazard Analysis and then developing a System Hazard Analysis) still applies but is built into the processes.

Preliminary safety analysis is effectively done through the assessments of the planned changes and the discussions on allocations to subsystems and deployments. Safety and operational analysis are then performed for each change identified as requiring analysis (in the form of an Operational Safety Hazard Analysis, Functional Failure Analysis, Interface Hazard Analysis, as applicable). Changes are only included in a planned deployment if they can be analysed and assured within the required timeframe, else they are deferred until they are able to be completed properly.

The project relies on a master Hazard Log as the live parent document. All analysis updates feed into the Hazard Log (which manages the application conditions process and the trace from controls to requirements and verification/validation) and is the input into the Safety Case SFAIRP argument.

4.5 The Importance of System Validation

The project has a high reliance on integration and system testing. This strongly supported with a world class lab with simulators and benches. The approach taken is to manage early integration of updates with a focus on backwards compatibility when possible. The onsite testing also validates the real-world scenarios for the safety critical systems.

There is an emphasis on effective testing of new safety features (requirements traced to engineering controls) and a validation approach, not just did the function work but did it work as the user would want or expect it to, ensuring that all system requirements are properly validated for the mission critical system

There is also detailed analysis and monitoring of operational and safety issues to define priorities for addressing defects.

4.6 Robust Application Condition Management

A robust process is required to manage the large amount of application conditions (2000+ overall). These are used to highlight changes in behaviour of functions, to manage the introduction of new functions, to identify new supported baselines, as well as to refine requirements by delivering residual risk scope to the Operator. There is a cooperative, interactive approach taken, to ensure that the risks being identified are clearly understood and that the corresponding procedural mitigations required to address those risks can be readily put into place.

Hitachi works closely with the Operator to cancel old application conditions that no longer apply when a new update is introduced or modifications are made (700+).

Application conditions are also used to manage defects with safety or high operational impact between identification and rectification of defects across the planned deployments.

4.7 Open Relationship with the ISA

Building a strong and open relationship with the ISA is crucial. There is constant and open communication to advise the ISA of planned works and to share issues and concerns and the proposed plans to address them, a process that is fully supported by the Operator.

Early exposure to safety related updates per deployment, well in advance of the deployment, allows the ISA time to fully understand the changes and to address any concerns or issues in advance of the planned deployment, preventing delays and the risk of rework to address comments.

4.8 Teamwork

There are multiple departments working on this project (both in Hitachi and within the Operator). It is crucial to work together as one team in order to deliver safe and reliable system on time. This requires strong open communication between departments, with a strong focus on working together to identify and address priority or blocking issues, whilst still respecting the required independence and adhering to the applicable standards.

4.9 Workload Management

The continuous development requires careful workload management. The workload is well distributed across the year as incorporated in the roadmap. Of the planned (up to 8 drops) annual deployments, not all deployments are the same. These are carefully spaced out, for example including key safety heavy updates to one subsystem at the start of the year and including safety heavy updates to another subsystem at the end of the year, as well as interleaving safety heavy updates with less intensive updates (such as Comms.). This allows the project to flatten peaks, troughs and manage workloads for the different departments, and deadlines throughout the year.

5 Alignment with Agile, Lean and Evolutionary Prototyping

The case study described in Section 4 demonstrates that Agile, Lean and Evolutionary Prototyping principles can be successfully implemented within a SIL-rated operational railway environment when combined with strong configuration management, rigorous verification and validation, formal safety assurance and effective stakeholder engagement. Rather than replacing traditional railway processes, these methods were integrated into existing assurance activities to support continuous system evolution while maintaining safety and operational performance. This section describes in more detail, the alignment, of the Case Study, with the Agile, Lean and

Evolutionary Prototyping concepts summarised in Section 3.

5.1 Early and Continuous Engagement with Operator

Continuous stakeholder engagement is a central principle of Agile, Lean and Evolutionary Prototyping. Agile emphasises customer collaboration and frequent feedback, Lean focuses on understanding customer value and eliminating barriers between organisations, and Evolutionary Prototyping on working with the customer to determine and refine the requirements.

From the case study it can be seen that the project demonstrates this by maintaining a long-term roadmap with the Operator, jointly assessing proposed changes, exposing updates to the Operator early through laboratory and site testing, including operational and human-factors input throughout development, and through the Operator participation in:

1. Design definition, before proceeding with the implementation: roadmap planning, reviews, demonstrations,
2. System acceptance, after implementation: validation, customer acceptance testing and application condition management.

This is a direct implementation of the Agile principle of continuous collaboration, the Lean principle of ensuring value is defined by the customer, and the Evolutionary Prototyping principle of Active User Involvement.

5.2 Strong Configuration Management

Although configuration management is not explicitly identified as an Agile practice in Section 3, it directly supports Agile's requirement for frequent and reliable delivery and Lean's focus on avoiding waste, whereas Evolutionary Prototyping requires strong management and control in order to create good software whilst preventing scope creep. Strong baseline control therefore prevents deployment errors, reduces rework, avoids waste associated with incorrect configurations, supports repeatable deployment processes, and prevents scope creep.

5.3 Verification, Verification, Verification

The central principle of Agile, Lean and Evolutionary Prototyping specifically emphasises continuous testing, frequent integration, early defect detection, and verification throughout the lifecycle.

The case study shows that the project's approach of embedding verification activities from change identification through implementation and deployment shows the embodiments of these principles. In particular, the attitude of "there is no time to correct later in the lifecycle" is particularly aligned with Lean thinking, which focuses on preventing defects rather than correcting them later, and the verification checks are crucial to the Evolutionary Prototyping process of initial requirements being refined through ongoing stakeholder feedback, and the discovery and clarification of requirements over time, in order to refine the prototype.

5.4 Safety Analysis

The Agile focus on Technical excellence and engineering discipline, the Lean focus on eliminating waste and minimising rework and the Evolutionary Prototyping focus on Incremental Delivery is aligned with the safety process described in the case study, where the traditional safety processes remain, but are tailored, refined and built into the overall process by integrating safety analysis into every planned change, maintaining a master hazard log, and preserving traceability from hazards through controls and verification. This also demonstrates the balance between Agile responsiveness and railway assurance discipline described in the literature review.

5.5 The Importance of System Validation

The central principle of Agile, Lean and Evolutionary Prototyping specifically emphasises continuous testing, frequent integration, validation throughout the lifecycle, and early feedback loops. The extensive use of simulation, integration testing and user-focused validation embodies these concepts. The case study shows that the validation in the project explicitly considers not only whether the function works but whether it meets operational user expectations. This is highly consistent with Agile's customer-value focus, Lean's emphasis on delivering real customer value, and Evolutionary Prototyping concept of validation that the system meets user expectations, through Incremental development and delivery.

5.6 Robust Application Condition Management

Application conditions support structured change management across operational constraints and assumptions. This supports Agile by enabling safe incremental changes, supports Lean by reducing ambiguity, avoiding duplicated analysis, and preventing operational confusion, and support Evolutionary Prototyping through management of the evolving system with continuous customer involvement.

5.7 Open Relationship with the ISA

The case study shows that the project's continuous engagement with the Independent Safety Assessor creates transparency, provides early feedback, identifies concerns before deployment, reduces potential rework, all of which are directly aligned with Agile Lean, and Evolutionary Prototyping concepts of early stakeholder involvement and the reduction of late lifecycle surprises. The traditional stakeholders in Agile, Lean, and Evolutionary Prototyping are the Operator (end-users), effectively, in this project we have treated the ISA as a stakeholder.

5.8 Teamwork

The case study shows the project's emphasis on cross-functional cooperation, shared problem solving, joint management of priorities and working as one team across organisations. This directly aligns with Agile's collaborative culture, Lean's goal of reducing organisational barriers and handoff waste, and Evolutionary Prototyping concept of active user involvement. It should be noted that this collaboration happens within the framework of the applicable railway standards, respecting at all times, the required

independence, and with a shared focus on safety and reliability.

5.9 Workload Management

The case study describes how the deployment roadmap intentionally smooths workloads across the year, balancing safety-intensive and less-intensive changes. This reduces bottlenecks, avoids overload, and improves overall process flow. This is very consistent with classic Lean flow optimisation principles, and Agile's principle to reduce the scope to what can be achieved in a sprint.

6 Future Ideas

The successful application of continuous development principles to a live operational railway has demonstrated that traditional railway assurance processes can be effectively tailored to support frequent deployments without compromising safety. As railway systems become increasingly digital, automated and data-driven, there are several opportunities to further improve the efficiency, effectiveness and responsiveness of continuous development and maintenance activities. These opportunities build upon the lessons learned from the case study and incorporate concepts from modern systems engineering, Agile development and Lean management approaches;

1. Increased adoption of digital engineering and model-based approaches across the entire system lifecycle.
2. Future enhancements could include automated scenario execution, continuous integration testing, automated requirements traceability checks and automated generation of verification evidence.
3. Additional detailed analysis of operational events, maintenance trends, system performance metrics and user feedback could provide valuable insight into emerging risks, operational bottlenecks and opportunities for improvement.

7 Conclusion

This paper has considered the challenges in the continuous development and maintenance of an operational safety critical system.

The case study described in this paper, shows that the project has successfully applied the core principles of Agile, Lean and Evolutionary Prototyping while retaining the governance and assurance required for a SIL-rated operational railway system. The key aspects of these methodologies, that have been tailored by the project, to ensure that they are still within the constraints of the applicable standards are:

1. Continuous stakeholder engagement and collaboration.
2. Frequent and incremental delivery of change.
3. Early verification and validation.
4. Continuous hazard management.
5. Defect prevention rather than defect correction.
6. Elimination of rework through communication and transparency.
7. Optimisation of workflow and deployment cadence.
8. Cross-functional teamwork and shared ownership.

Applying this case study approach to other railway projects like multi-operator and passenger railway environments introduces an additional dimension of organisational and operational complexity. Changes may affect several infrastructure managers, train operators, passenger service organisations, control centres, maintainers, regulators, emergency services, workforce representatives and end-user groups, each with different operating rules, service obligations, risk tolerances and implementation constraints. In such environments, stakeholder consultation cannot be treated as a single approval activity at the end of development. It should be planned as a continuous, proportionate and traceable workstream within each increment, beginning with the definition of the operational need and continuing through impact assessment, option selection, requirements development, simulation, trial operation, training, procedure updates, readiness assessment and post-deployment review. A practical conclusion is that this case study development is transferable to these broader railway contexts when the roadmap is jointly governed, consultation is embedded in the development cycle, affected parties are engaged early enough to influence the solution, and deployment decisions remain based on verified safety evidence and demonstrable operational readiness.

Overall, this case study can be viewed as a practical implementation of Agile, Lean and Evolutionary Prototyping concepts adapted for a continuously evolving operational safety-critical railway system, demonstrating that Agile responsiveness, Lean efficiency, and Evolutionary Prototyping continuous evolution, can coexist with formal railway safety assurance processes. Hence the key contributions and additions to current learning discussed in this paper were a consideration of a case study functions, which considered how:

1. Advancement of knowledge of applying multiple methodologies within the same development lifecycle for a railway project
2. Improved understanding of how to effectively and efficiently tailor the standard development lifecycle for a railway project in practice, noting that things can still go wrong (and will!) but the processes can then be used to quickly assess the impact and plan the required mitigation tasks and timelines
3. Sharing experience from use of such a tailored development lifecycle in an industrial case study.

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